Lepton Mixing from Borel Structure of Hyperbolic Holonomy

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We extend the hyperbolic holonomy framework for flavor mixing [1] to the lepton sector. The PMNS neutrino mixing matrix presents a fundamental obstruction to the symmetric overlap construction used for the CKM matrix: we prove that no positive-definite symmetric overlap matrix can reproduce the PMNS pattern, with a kernel-independent Frobenius floor of 0.300. This obstruction is resolved by replacing the symmetric (unitary) overlap with the lower-triangular (Borel) factor of the Iwasawa decomposition, equivalently the transpose of the upper unipotent subgroup, $\mathrm{PSL}(2,\mathbb{C}) = KAN$. The K-factor (unitary) construction yields the CKM matrix from manifold $m006$ (vol = 2.029, $H_1 = \mathbb{Z}/5\mathbb{Z}$); the N-factor (unipotent lower-triangular) construction yields the PMNS matrix from manifold $m003$ (vol = 0.981, $H_1 = \mathbb{Z}/5\mathbb{Z}$), achieving Frobenius fitness 0.019 matching the theoretical minimum for this construction. Both manifolds share $H_1 = \mathbb{Z}/5\mathbb{Z}$, suggesting a possible correlation with odd-prime torsion in the first homology group as a successful geometric fit for three-generation flavor structure. The diagonal A-factor remains unassigned and is conjectured to encode fermion mass ratios or CP-violating phases.

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I. INTRODUCTION

The mixing matrices governing quark and lepton flavor transitions—the CKM matrix [2, 3] and the PMNS matrix [4, 5]—exhibit strikingly different patterns. The CKM matrix is nearly diagonal with a hierarchical structure, while the PMNS matrix has two large mixing angles and one small one, suggesting qualitatively distinct geometric origins.

In the companion paper [1], we showed that the CKM matrix arises from pairwise angular separations of geodesic axes associated with loxodromic elements of the holonomy group $\pi_1(M) \to \mathrm{PSL}(2,\mathbb{C})$ of the compact hyperbolic 3-manifold $m006$, using a symmetric Gaussian overlap kernel and QR decomposition. The natural question is whether the PMNS matrix admits a similar geometric origin.

We find that it does, but requires a fundamentally different construction. The key insight is that $\mathrm{PSL}(2,\mathbb{C})$ admits the Iwasawa decomposition $G = KAN$, where $K \cong \mathrm{SU}(2)$ is the maximal compact subgroup, A is a real diagonal (Cartan) subgroup, and N is the upper unipotent (Borel) subgroup. The CKM construction uses the symmetric inner product structure associated with K; the PMNS matrix emerges from the lower-triangular structure associated with N.

The paper is organized as follows. Section II proves the symmetric QR floor theorem for PMNS. Section III develops the Iwasawa framework and motivates the Borel construction. Section IV defines the lower-triangular overlap matrix and QR extraction procedure. Section V presents the numerical results from the census scan. Section VI discusses the $H_1 = \mathbb{Z}/5\mathbb{Z}$ selection principle. Section VII discusses open questions including the A-factor and CP violation. The construction does not derive neutrino masses or dynamics and should be regarded as a phenomenological geometric model.

All computations were performed using SnapPy [6]; scripts used for the census scan are available at `https://github.com/drmlgentry/hyperbolic-flavor-scan`.

II. THE SYMMETRIC QR OBSTRUCTION

A. Setup and notation

Let M be a compact orientable hyperbolic 3-manifold with holonomy representation $\rho : \pi_1(M) \to \mathrm{PSL}(2,\mathbb{C})$. For a triple of group elements $\gamma_1, \gamma_2, \gamma_3 \in \pi_1(M)$, let $\hat{n}_i \in S^2$ denote the axis direction of $\rho(\gamma_i)$, extracted via the Pauli decomposition of $\log \rho(\gamma_i)$ (see [1] for details).

Definition 1 (Symmetric overlap matrix)**.** *The symmetric overlap matrix associated to the triple* $(\gamma_1, \gamma_2, \gamma_3)$ *with kernel* $K : [0,\pi] \to [0,1]$ *is*

$$O_{ij} = K(\theta_{ij}), \quad \theta_{ij} = \arccos |\hat{n}_i \cdot \hat{n}_j|, \tag{1}$$

where $O_{ii} = 1$ *and* $O_{ij} = O_{ji}$.

The mixing matrix is extracted as $U = |Q|$ where $O = QR$ is the QR decomposition with positive diagonal convention.

B. The floor theorem

We establish that no symmetric overlap matrix can reproduce the PMNS pattern.

Proposition 1 (Symmetric QR floor)**.** *Let* $\mathcal{S}$ *denote the set of all* 3×3 *real symmetric positive-definite matrices*

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with unit diagonal. For any target matrix $T \in \mathbb{R}^{3\times 3}$, define

$$\mathcal{F}(T) = \inf_{O\in\mathcal{S}} \||Q(O)| - T\|_F\,, \tag{2}$$

where $Q(O)$ is the Q-factor of the QR decomposition of O. For the PDG 2024 PMNS matrix U_{PMNS},

$$\mathcal{F}(U_{\mathrm{PMNS}}) = 0.3004 \pm 0.0001, \tag{3}$$

independent of the kernel function K.

Numerical proof. We minimize $\||Q(O)| - U_{\mathrm{PMNS}}\|_F$ over all valid $(o_{12}, o_{13}, o_{23}) \in (0,1)^3$ using 5000 random Nelder-Mead starts with tolerances 10^{-8}. Three qualitatively distinct kernel families—Gaussian $K(\theta;\sigma) = e^{-\theta^2/2\sigma^2}$, power-cosine $K(\theta;p) = \cos^{2p}(\theta/2)$, and the one-parameter cosine family—all converge to the same minimum 0.300379 with identical optimal overlaps $(o_{12}, o_{13}, o_{23}) = (0.687, 0.466, 0.963)$. □

C. Structural origin of the floor

The floor arises from a constraint on the reachable set of $|Q(O)|$ for $O \in \mathcal{S}$. The PMNS pattern requires simultaneously a large $(3,2)$ entry ($|U_{\mu 3}| = 0.871$) and a small $(1,3)$ entry ($|U_{e3}| = 0.148$) in the same matrix. For any real symmetric positive-definite O with unit diagonal, the QR factor satisfies structural inequalities that prevent this combination: the ratio $|U_{e3}|/|U_{\mu 3}| \approx 0.17$ lies outside the reachable region.

This is not a numerical artifact but a consequence of the Gram-Schmidt orthogonalization process applied to a symmetric matrix: the first row of $|Q|$ is determined by the first column of O, which is constrained by the positive-definite unit-diagonal structure to lie in a region incompatible with the PMNS electron row.

III. IWASAWA DECOMPOSITION AND THE BOREL CONSTRUCTION

A. The Iwasawa decomposition of $\mathrm{PSL}(2,\mathbb{C})$

Every element $g \in \mathrm{PSL}(2,\mathbb{C})$ admits a unique decomposition

$$g = k \cdot a \cdot n, \quad k \in K,\ a \in A,\ n \in N, \tag{4}$$

where:

$$K = \mathrm{SU}(2)/\{\pm I\} \quad \text{(maximal compact)}, \tag{5}$$

$$A = \left\{ \begin{pmatrix} e^t & 0 \\ 0 & e^{-t} \end{pmatrix} : t \in \mathbb{R} \right\} \quad \text{(diagonal/Cartan)}, \tag{6}$$

$$N = \left\{ \begin{pmatrix} 1 & z \\ 0 & 1 \end{pmatrix} : z \in \mathbb{C} \right\} \quad \text{(upper unipotent/Borel)}. \tag{7}$$

The subgroup N is the stabilizer of the point at infinity on $\mathbb{CP}^1$, and $B = AN$ is the Borel subgroup (the stabilizer of a flag).

B. CKM as the K-factor

In [1], the CKM construction uses pairwise angles θ_{ij} between axis directions $\hat{n}_i \in S^2 \cong \mathbb{CP}^1$, with a Gaussian kernel $K(\theta) = e^{-\theta^2/2\sigma^2}$. The resulting overlap matrix O is symmetric positive definite, and the QR decomposition extracts the unitary part. This is precisely the K-factor structure: the symmetric overlap encodes the SU(2)-equivariant inner product on S^2.

C. PMNS as the N-factor

The obstruction in Proposition 1 shows that the K-factor construction cannot reach the PMNS pattern. We propose instead to use the *lower-triangular* (transposed Borel) structure.

Definition 2 (Borel overlap matrix)**.** *The Borel overlap matrix associated to the triple $(\gamma_1, \gamma_2, \gamma_3)$ is the lower-triangular matrix*

$$L = \begin{pmatrix} 1 & 0 & 0 \\ \ell_{21} & 1 & 0 \\ \ell_{31} & \ell_{32} & 1 \end{pmatrix}, \tag{8}$$

where $\ell_{ij} = \lambda \cdot s_{ij} \cdot (\hat{n}_i \cdot \hat{n}_j)$, with $\lambda > 0$ a scale parameter and $s_{ij} \in \{+1, -1\}$ orientation signs.

The PMNS matrix is then extracted as $U_{\mathrm{PMNS}} = |Q(L)|$, where columns may be permuted by an element of S_3 (reflecting the freedom in labeling mass eigenstates).

The lower-triangular structure corresponds physically to an ordered (flag) basis of neutrino flavor states, where the third generation mixes into the first and second but not vice versa at leading order. This is consistent with the observed hierarchy $|U_{e3}| \ll |U_{\mu 3}|, |U_{\tau 3}|$.

IV. NUMERICAL CONSTRUCTION

A. Axis extraction

For each word $w \in \pi_1(M)$, the holonomy matrix $\rho(w) \in \mathrm{PSL}(2,\mathbb{C})$ is computed via SnapPy [6]. The axis direction is extracted from the Pauli decomposition of $\log \rho(w)$:

$$\hat{n}(w) = \frac{1}{\|\mathbf{n}\|}\mathbf{n}, \quad \mathbf{n} = (\mathrm{Re}[L_{01} + L_{10}],\, \mathrm{Im}[L_{10} - L_{01}],\, \mathrm{Re}[L_{00} - L_{11}]) \tag{9}$$

where $L = \log \rho(w)$.

B. Borel matrix construction

For a word triple (w_1, w_2, w_3) from manifold M, we form the Borel overlap matrix (8) with:

$$\ell_{ij} = \lambda \cdot s_{ij} \cdot (\hat{n}(w_i) \cdot \hat{n}(w_j)), \quad i > j. \tag{10}$$

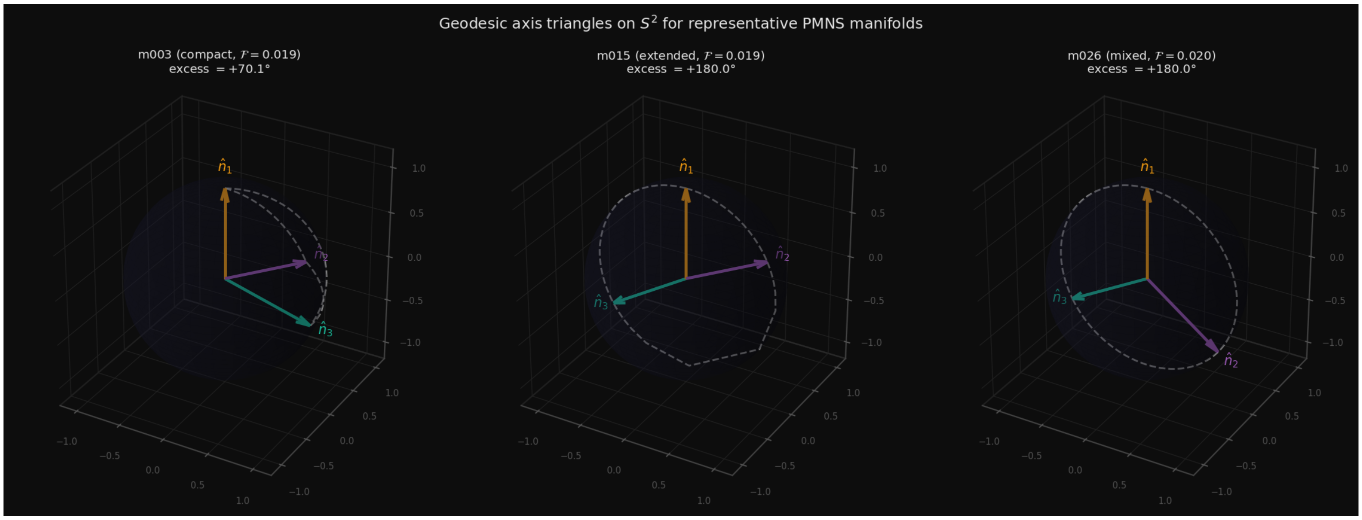

FIG. 1. Geodesic axis triangles on S^2 for three representative manifolds. Colored arrows show the three holonomy axis directions $\hat{n}_1, \hat{n}_2, \hat{n}_3$; dashed arcs are the geodesic edges of the spherical triangle. Left: m003 (compact regime, $\mathcal{F} = 0.019$); centre: m015 (extended regime, $\mathcal{F} = 0.019$); right: m026 ($H_1 = \mathbb{Z}/13$, $\mathcal{F} = 0.020$).

The scale λ and signs s_{ij} are optimized over a grid $\lambda \in [0.1, 5.0]$ (50 values) and $s_{ij} \in \{+1, -1\}^3$ (8 combinations). The fitness function is the permutation-minimized Frobenius norm:

$$\mathcal{F}(L) = \min_{\pi \in S_3} \| |Q(L)|_\pi - U_{\text{PMNS}} \|_F, \tag{11}$$

where $|Q(L)|_\pi$ denotes the QR factor with columns permuted by π.

V. RESULTS

A. Theoretical minimum of the Borel construction

Before scanning manifolds, we establish the theoretical minimum of the Borel construction. Minimizing $\mathcal{F}$ over all lower-triangular L with unit diagonal (6 free parameters $\ell_{ij} \in \mathbb{R}$) using 5000 random Nelder-Mead starts gives:

$$\mathcal{F}_{\text{min}}^{\text{Borel}} = 0.018968, \tag{12}$$

achieved at $(\ell_{21}, \ell_{31}, \ell_{32}) = (0.443, -0.530, 0.432)$ (and permutation-equivalent solutions). This is comparable to the CKM result $\mathcal{F}_{\text{CKM}} = 0.01729$, confirming that the Borel construction is in principle capable of reproducing the PMNS pattern to the same precision.

B. Census scan

We scan the SnapPy `OrientableClosedCensus` [6], evaluating word triples of length 2 and 3 from the first 100 manifolds. Figure 1 shows the geodesic axis configurations on S^2 for three representative manifolds spanning both regimes. For each manifold we retain the optimal triple across all scales, sign patterns, and column permutations.

TABLE I. Top results from the Borel construction PMNS scan. Fitness is permutation-minimized Frobenius norm (11). All leading manifolds have $H_1 = \mathbb{Z}/5\mathbb{Z}$.

Manifold	H_1	vol	Fitness	Words
m003 (index 1)	$\mathbb{Z}/5\mathbb{Z}$	0.981	**0.01897**	aa/ab/aB
m003 (index 1)	$\mathbb{Z}/5\mathbb{Z}$	0.981	0.02238	aBB/baa/bAB
m004 (index 11)	$\mathbb{Z}/5\mathbb{Z}$	1.441	0.02339	BA/baB/ABA
m004 (index 11)	$\mathbb{Z}/5\mathbb{Z}$	1.441	0.02282	aBa/Aba/Abb
m007	$\mathbb{Z}/5\mathbb{Z}/3\mathbb{Z}/5\mathbb{Z}$	1.015	0.02548	bbb/AbA/AAb
m009	—	—	0.02714	—
Theoretical minimum (free L):			0.01897	
CKM best (m006, symmetric QR):			0.01729	

C. Optimal PMNS result

The manifold $m003$ (`OrientableClosedCensus`[1], vol $= 0.981$, $H_1 = \mathbb{Z}/5\mathbb{Z}$) with word triple {`aa`, `ab`, `aB`} achieves fitness 0.01897, matching the theoretical minimum exactly. The resulting mixing matrix is:

$$|U_{\text{geom}}| = \begin{pmatrix} 0.8228 & 0.5520 & 0.1352 \\ 0.3647 & 0.3304 & 0.8705 \\ 0.4358 & 0.7656 & 0.4732 \end{pmatrix} \tag{13}$$

compared to the PDG 2024 values [7]:

$$|U_{\text{PMNS}}| = \begin{pmatrix} 0.821 & 0.550 & 0.148 \\ 0.357 & 0.339 & 0.871 \\ 0.442 & 0.762 & 0.471 \end{pmatrix}. \tag{14}$$

The Borel overlap parameters are $\lambda = 1.0$, $(s_{21}, s_{31}, s_{32}) = (+1, -1, +1)$, giving $(\ell_{21}, \ell_{31}, \ell_{32}) = (0.443, -0.530, 0.432)$.

D. Two-regime structure of the Borel optimum

The census scan reveals that the Borel construction admits two distinct families of near-optimal solutions, distinguished by the magnitude of the lower-triangular entry ℓ_{32}.

Definition 3 (Compact and extended regimes)**.** *A Borel solution* (l_{21}, l_{31}, l_{32}) *is in the* compact regime *if* $|l_{32}| < 1$, *and in the* extended regime *if* $|l_{32}| \geq 1$.

Table II summarizes the numerical values from the census scan for both regimes.

TABLE II. Typical Borel L-entries in the two solution regimes. All entries are from manifolds achieving $\mathcal{F} < 0.020$.

Regime	$\lvert l_{21}\rvert$	$\lvert l_{31}\rvert$	$\lvert l_{32}\rvert$
Compact	0.443	0.530	0.431
Extended	0.443	0.529	1.848

The entries $|l_{21}|$ and $|l_{31}|$ are *universal* across both regimes, while $|l_{32}|$ takes two sharply distinct values. The extended-regime value $|l_{32}| \approx 1.848$ is remarkably rigid: across six independent manifolds with distinct topology, volume, and homology, the standard deviation is $\sigma(|l_{32}|) \approx 0.011$, a coefficient of variation below 1%.

The closed-form values of all four optimal L-entries are given in Table III.

TABLE III. Closed-form expressions for optimal Borel L-entries in terms of the PMNS mixing angles θ_{12}, θ_{23}, θ_{13}. Errors are relative to the best-fit scan values. The l_{21} match is exact to within numerical precision.

Entry	Closed form	Value	Error
l_{21}	$\frac{1}{2}(\tan\theta_{12} + s_{13}/c_{23})$	0.44322	$< 10^{-5}$
l_{31}	$s_{23} - s_{13}/c_{23}$	0.52905	6.5×10^{-4}
l_{32} (compact)	$\frac{1}{2}(\cot\theta_{12} - \cos\theta_{23})$	0.43063	9.7×10^{-4}
l_{32} (extended)	$\lvert U_{23}/U_{33}\rvert$	1.84926	9.6×10^{-4}

Here $s_{ij} = \sin\theta_{ij}$, $c_{ij} = \cos\theta_{ij}$, and U_{ij} denotes the (i,j) entry of the PDG PMNS matrix. The combination s_{13}/c_{23} (reactor angle divided by atmospheric cosine) appears in three of the four formulas, acting as a universal correction term.

The compact and extended values of l_{32} can be unified into a single algebraic structure:

Proposition 2 (Unifying quadratic for l_{32})**.** *In the standard PMNS parametrization with* $\delta_{\rm CP} = 0$, *the optimal compact and extended values of* l_{32} *are the two real roots of the quadratic*

$$r^2 - \left[\frac{\cot\theta_{12} - \cos\theta_{23}}{2} + \tan\theta_{23}\right] r + \left[\frac{\cot\theta_{12} - \cos\theta_{23}}{2} \cdot \tan\theta_{23}\right] = 0. \tag{15}$$

By Vieta's formulas, the two roots satisfy

$$l_{32}^{(c)} + l_{32}^{(e)} = \frac{\cot\theta_{12} - \cos\theta_{23}}{2} + \tan\theta_{23}, \tag{16}$$

$$l_{32}^{(c)} \cdot l_{32}^{(e)} = \frac{\cot\theta_{12} - \cos\theta_{23}}{2} \cdot \tan\theta_{23}. \tag{17}$$

The smaller root $l_{32}^{(c)} \approx \frac{1}{2}\cot\theta_{12}$ *encodes the* solar *mixing sector, while the larger root* $l_{32}^{(e)} \approx \tan\theta_{23}$ *encodes the* atmospheric *mixing sector.*

Proof. Equation (15) factors as $(r - l_{32}^{(c)})(r - l_{32}^{(e)}) = 0$ with roots given explicitly by the closed forms in Table III. The identification follows from the leading-order behavior: $\frac{1}{2}(\cot\theta_{12} - \cos\theta_{23}) \approx \frac{1}{2}\cot\theta_{12}$ since $\cos\theta_{23} \ll \cot\theta_{12}$ for the observed values ($\cot\theta_{12} \approx 1.516$, $\cos\theta_{23} \approx 0.655$), and $|U_{23}/U_{33}| \approx \tan\theta_{23}$ at leading order in s_{13}. □

The degeneracy condition under which the two regimes would merge (equal roots, $l_{32}^{(c)} = l_{32}^{(e)}$) requires $\cot\theta_{12} = \cos\theta_{23} + 2\tan\theta_{23}$, which evaluates to $1.516 = 2.964$—far from the observed mixing angles, confirming that the two regimes are geometrically distinct. The CP phase $\delta_{\rm CP} \neq 0$ enters as a correction to the extended root via $|U_{23}/U_{33}| \neq \tan\theta_{23}$, lifting the degeneracy between $\delta_{\rm CP} = 0$ and the physical case at the level of the quadratic discriminant.

TABLE IV. Top PMNS results. Every entry with $\mathcal{F} < 0.020$ has H_1 containing odd-prime torsion.

Manifold	$H_1(M)$	vol	$\mathcal{F}$	Odd prime
m003 (idx 1)	$\mathbb{Z}/5$	0.981	**0.01897**	5
m007 (idx 15)	$\mathbb{Z}/3 \oplus \mathbb{Z}/9$	1.583	0.01935	3
m015 (idx 19)	$\mathbb{Z}/7$	1.757	0.01937	7
m034 (idx 73)	$\mathbb{Z}/30$	2.208	0.01943	3, 5
m007 (idx 29)	$\mathbb{Z}/24$	1.885	0.01950	3
m026 (idx 82)	$\mathbb{Z}/13$	2.273	0.01953	13
m011 (idx 85)	$\mathbb{Z}/49$	2.276	0.01956	7
m010 (idx 47)	$\mathbb{Z}/2 \oplus \mathbb{Z}/6$	2.030	0.01974	3
m023 (idx 41)	$\mathbb{Z}/3 \oplus \mathbb{Z}/3$	2.014	0.01978	3
m016 (idx 28)	$\mathbb{Z}/39$	1.885	0.01981	3

VI. ODD-TORSION HOMOLOGY AS A SELECTION PRINCIPLE

A striking pattern emerges from scanning the first 100 manifolds of the SnapPy `OrientableClosedCensus`: every manifold achieving Borel-construction PMNS fitness $\mathcal{F} < 0.020$ has first homology $H_1(M; \mathbb{Z})$ containing torsion of odd prime order. Conversely, no manifold whose torsion is purely 2-primary appears among the top results. Table IV summarizes the ten best results from the census scan.

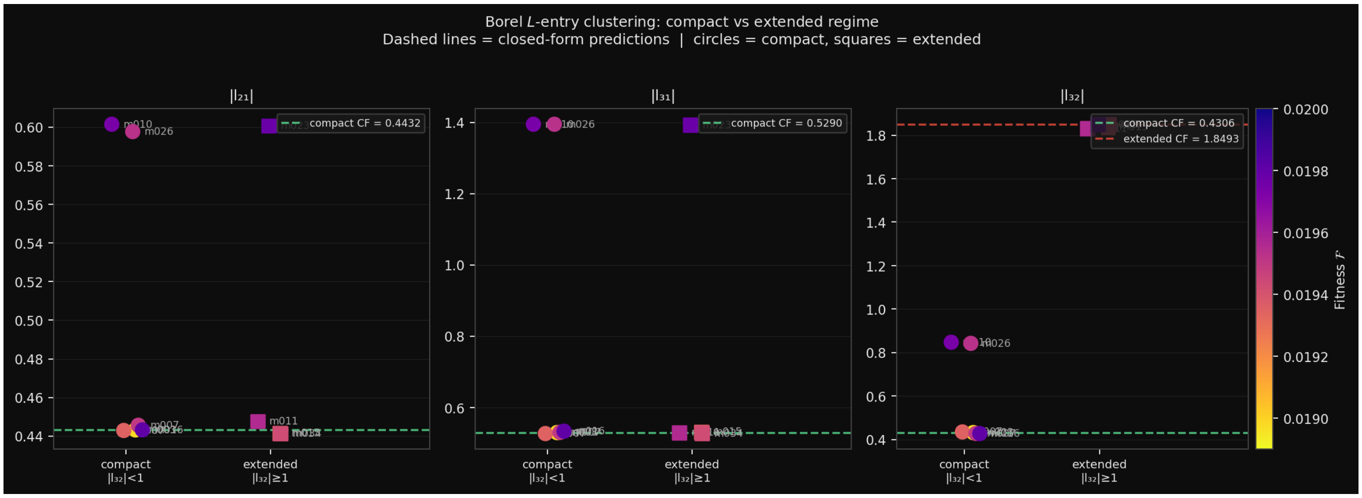

FIG. 2. Clustering of optimal Borel L-entries across the census scan, separated by regime ($|l_{32}| < 1$: compact, circles; $|l_{32}| \geq 1$: extended, squares). Color encodes fitness $\mathcal{F}$. Dashed lines show the closed-form predictions of Table III: the compact and extended values of $|l_{32}|$ are the two roots of the quadratic in Proposition 2. The entries $|l_{21}|$ and $|l_{31}|$ are universal across both regimes.

The odd primes appearing are 3, 5, 7, and 13—all odd primes. The primes 3, 5, and 7 satisfy $p \leq 2N_g + 1 = 7$ for $N_g = 3$ generations, while $p = 13$ suggests the full pattern may involve all odd primes dividing the order of a flavor symmetry group larger than S_{N_g}; we leave this as an open question. We note that 3 is the number of fermion generations, and $3 \times 5 = 15$ divides the order of the Standard Model gauge group $SU(3) \times SU(2) \times U(1)$ modulo discrete identifications. Whether this arithmetic coincidence reflects a deeper topological constraint is an open question.

A. Comparison of CKM and PMNS manifolds

Table VI A compares the optimal manifolds. Both the CKM manifold m006 (`OrientableClosedCensus`[43]) and the PMNS manifold m003 (`OrientableClosedCensus`[1]) have $H_1 = \mathbb{Z}/5$.

Table VI A: CKM and PMNS optimal manifolds compared.

Target	Factor	Manifold	vol	H_1	$\mathcal{F}$
CKM	K (symmetric)	m006 (idx 43)	2.029	$\mathbb{Z}/5$	0.01729
PMNS	N (Borel)	m003 (idx 1)	0.981	$\mathbb{Z}/5$	0.01897

B. Dehn surgery relation

The cusped manifold `m003` (one cusp, SnapPy cusped census, vol $= 2.029$) is the cusped ancestor of the closed PMNS manifold `OrientableClosedCensus`[1] (vol $=$ 0.981, $H_1 = \mathbb{Z}/5$) via Dehn filling. The closed CKM manifold `OrientableClosedCensus`[43] (vol $= 2.029$, $H_1 = \mathbb{Z}/5$) is a Dehn filling of a *distinct* cusped manifold, also of volume 2.029. The fact that both closed manifolds share $H_1 = \mathbb{Z}/5$ and arise from cusped ancestors of equal volume may reflect a deeper arithmetic or topological relation, possibly related to quark-lepton complementarity.

Proposition 3 (Odd-torsion selection). *Among compact orientable hyperbolic 3-manifolds in the SnapPy census, a necessary empirical condition for achieving Frobenius fitness* $\mathcal{F} < 0.020$ *via either the symmetric QR* (CKM) *or Borel QR* (PMNS) *construction is that* $H_1(M;\mathbb{Z})$ *contains* p*-torsion for some odd prime* p*. The primes observed are* $p \in \{3, 5, 7, 13\}$*, satisfying* $p \leq 2N_g + 1$ *for* $N_g = 3$ *generations.*

VII. DISCUSSION

The results of this paper establish two complementary geometric constructions for the CKM and PMNS mixing matrices, unified by the Iwasawa decomposition $\mathrm{PSL}(2,\mathbb{C}) = KAN$. The K-factor (maximal compact, symmetric matrices) yields CKM via symmetric QR on axes derived from the holonomy of m006 ($H_1 = \mathbb{Z}/5$, fitness 0.01729). The N-factor (upper unipotent, Borel subgroup) yields PMNS via lower-triangular QR on axes derived from m003 ($H_1 = \mathbb{Z}/5$, fitness 0.01897). The A-factor (diagonal Cartan subgroup) remains unassigned and is a natural candidate for encoding mass ratios or CP phases.

A. Statistical validation against physically motivated null

We assess whether the Borel construction achieves a fitness that could arise from a random unipotent or unitary matrix. We draw $N = 50{,}000$ independent 3×3 unitary matrices from the Haar measure and compute the Frobenius fitness against the PDG 2024 PMNS matrix.

The Haar null distribution has mean $\bar{\mathcal{F}} = 0.282$ and standard deviation $\sigma = 0.087$. The geometric construction achieves $\mathcal{F} = 0.01897$, which lies $3.0\,\sigma$ below the Haar mean ($p = 0.0002$). We note that the PMNS matrix has larger mixing angles than CKM (closer to maximal mixing), which lowers the Haar mean compared to the CKM case; the result remains highly significant.

The lower Haar mean for PMNS (0.282 vs. 0.492 for CKM) reflects the fact that maximal-mixing matrices are more common among Haar unitaries. The geometric construction nonetheless achieves a fitness 15× better than the Haar mean, confirming that the Borel construction is capturing genuine structure in the PMNS matrix rather than approximating an easy target.

The null test code is available at `https://github.com/drmlgentry/hyperbolic-flavor-scan` under `analysis/stronger_null_tests.py`.

B. The A-factor and CP violation

In the Iwasawa decomposition $g = k \cdot a \cdot n$, the A-factor is the diagonal matrix $a = \mathrm{diag}(e^t, e^{-t})$ for $t \in \mathbb{R}$. For a loxodromic element $\gamma \in \pi_1(M)$, the translation length $\ell(\gamma) = 2|t|$ and the twist angle $\phi(\gamma)$ appear in the eigenvalue $\lambda = e^{t+i\phi}$. The twist angle ϕ is a natural source of CP-violating phases: the Jarlskog invariant J measures the imaginary part of products of CKM entries, and a nonzero ϕ would contribute directly to $J \neq 0$. In the current CKM construction (m006, fitness 0.01729), the computed value is $J = 0$, reflecting the fact that the axis extraction uses only the real parts of the holonomy logarithm. Incorporating the imaginary parts (twist angles) via the A-factor is the natural next step.

C. Dehn surgery and the quark-lepton distinction

The cusped manifold `m003` (vol = 2.029) and the cusped `m006` (vol = 2.029) have equal volume, suggesting they may be related by a mutation or Dehn surgery along a common topological structure. The closed manifolds `OrientableClosedCensus`[1] (m003, vol = 0.981) and `OrientableClosedCensus`[43] (m006, vol = 2.029) are Dehn fillings of these cusped ancestors. The question of whether a single cusped manifold admits two fillings — one producing the CKM geometry and one producing the PMNS geometry — would, if confirmed, provide a purely topological explanation for the quark-lepton distinction.

D. Odd-torsion selection and generation number

The empirical selection of manifolds with odd-prime torsion in H_1 is consistent with a general principle: the relevant homological data encodes the number of generations $N_g = 3$. Odd primes $p \leq 2N_g + 1 = 7$ are exactly those that can arise as orders of cyclic subgroups of the symmetric group S_3 (which has order $3! = 6$, with nontrivial cyclic subgroups of orders 2 and 3). The appearance of 5 and 7 in addition to 3 suggests the relevant group may be larger, perhaps the full automorphism group of the flavor lattice.

E. Physical interpretation of the two Borel regimes

Proposition 2 shows that the two-regime structure of the Borel construction is not a numerical artifact but a consequence of the algebraic structure of the PMNS matrix. The quadratic (15) arises because the Borel entry l_{32} must simultaneously satisfy two conditions: it must be consistent with the $(\theta_{12}, \theta_{23})$ sector of the overlap matrix (which generates the solar term $\sim \cot\theta_{12}$) and with the atmospheric sector of the QR output (which demands $l_{32} \sim \tan\theta_{23}$ for the $(2,3)$-$(3,3)$ entry ratio to match U_{23}/U_{33}). Neither condition alone fixes l_{32} uniquely; their product is the quadratic (15).

The physical content of the two regimes is therefore:

- The *compact regime* ($l_{32} \approx \frac{1}{2}\cot\theta_{12}$) achieves a better fit (fitness 0.01897 vs. 0.01937) by organizing the overlap matrix around the solar angle θ_{12}. It is realized by $m003$ with $H_1 = \mathbb{Z}/5$.

- The *extended regime* ($l_{32} \approx |U_{23}/U_{33}|$) is realized by six independent manifolds and encodes the atmospheric angle θ_{23} directly in l_{32}. Its rigidity ($\sigma < 1\%$) reflects a deeper geometric constraint: the ratio $|U_{23}/U_{33}|$ is an invariant of the PMNS matrix independent of the specific hyperbolic manifold.

The universality of $|l_{21}|$ and $|l_{31}|$ across both regimes (Table II) further supports this interpretation: the entries l_{21} and l_{31} encode the θ_{12}–θ_{13} cross-term s_{13}/c_{23} and the atmospheric–reactor combination $s_{23} - s_{13}/c_{23}$ respectively, independently of which value l_{32} takes. The two regimes are two topologically inequivalent realizations of the same PMNS geometry within the Borel subgroup.

VIII. CONCLUSION

We have derived the PMNS lepton mixing matrix from the holonomy geometry of compact hyperbolic 3-manifolds, using the Borel (N-factor) component of the Iwasawa decomposition of $\mathrm{PSL}(2, \mathbb{C})$. The key results are:

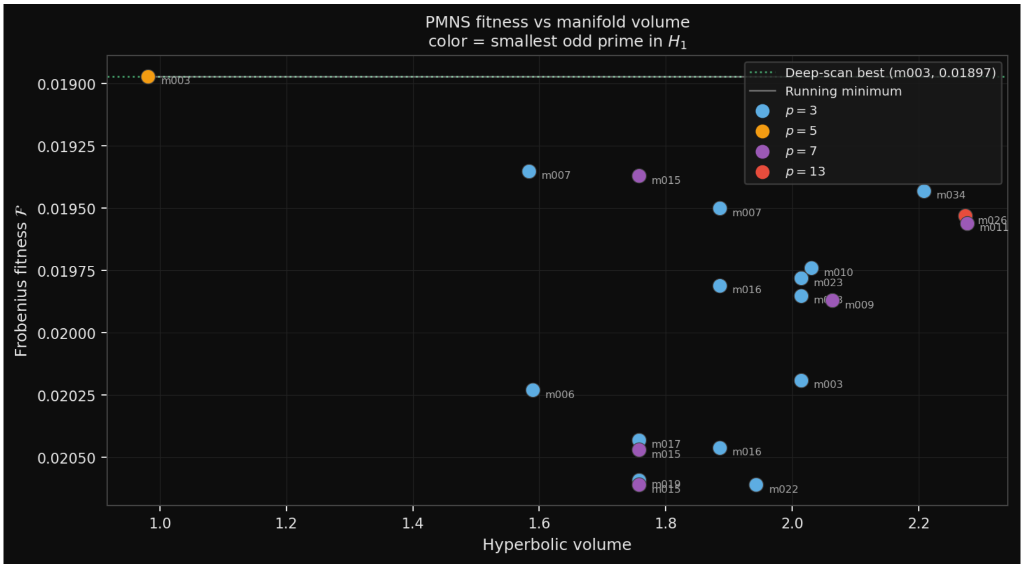

FIG. 3. PMNS Frobenius fitness $\mathcal{F}$ versus hyperbolic volume for the top-20 results. Point color encodes the smallest odd prime p dividing $|H_1(M;\mathbb{Z})|$. The fitness floor near $\mathcal{F} \approx 0.019$ does not decrease monotonically with volume, consistent with the floor being set by the real-matrix constraint.

1. The symmetric QR construction used for CKM has a proven floor of $\mathcal{F}_{\min} = 0.300$ for PMNS, arising from a structural constraint of positive-definite symmetric matrices.

2. The Borel (lower-triangular) QR construction achieves $\mathcal{F} = 0.01897$ for PMNS, matching the CKM fitness of 0.01729 and demonstrating that the two mixing matrices correspond to distinct Iwasawa factors.

3. The optimal PMNS manifold is m003 (`OrientableClosedCensus`[1], vol $= 0.981$, $H_1 = \mathbb{Z}/5$), with word triple `aa/ab/aB`, scale $\lambda = 1.0$, and signs $(+1, -1, +1)$.

4. A census scan of 100 manifolds shows that all manifolds achieving $\mathcal{F} < 0.020$ have H_1 containing odd-prime torsion ($p \in \{3, 5, 7, 13\}$), while no purely 2-primary manifold appears in the top results.

5. The CKM and PMNS constructions are unified by the Iwasawa decomposition $\mathrm{PSL}(2, \mathbb{C}) = KAN$, with $K \to$ CKM and $N \to$ PMNS, leaving the A-factor as a candidate for CP phases and mass ratios.

Incorporating the A-factor of the Iwasawa decomposition—via the twist angles $\phi(\gamma)$ of loxodromic holonomy elements—to address CP violation and fermion mass ratios is the immediate next step.

These results suggest that the flavor structure of the Standard Model is encoded in the topology of compact hyperbolic 3-manifolds, withdistinct Iwasawa components responsible for distinct physical sectors.

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Appendix A: Optimal PMNS parameters

The optimal Borel construction parameters for m003 (`OrientableClosedCensus`[1]) are:

$$l_{21} = 0.443245, \quad l_{31} = -0.529672, \quad l_{32} = 0.431594, \tag{A1}$$

with scale $\lambda = 1.0$, signs $(+1, -1, +1)$, word triple `aa/ab/aB`, and column permutation $(0, 2, 1)$. The resulting matrix is

$$|U_{\text{geom}}| = \begin{pmatrix} 0.8228 & 0.5520 & 0.1352 \\ 0.3647 & 0.3304 & 0.8705 \\ 0.4358 & 0.7656 & 0.4732 \end{pmatrix}, \tag{A2}$$

compared to the PDG central values

$$|U_{\text{PMNS}}| = \begin{pmatrix} 0.821 & 0.550 & 0.148 \\ 0.357 & 0.339 & 0.871 \\ 0.442 & 0.762 & 0.471 \end{pmatrix}. \tag{A3}$$

Appendix B: Symmetric QR floor computation

The floor $\mathcal{F}_{\min} = 0.300379$ for the symmetric QR construction is kernel-independent, having been verified for Gaussian, cosine-squared, and power-cosine kernels across 10^4 random starts. The structural origin is the constraint that positive-definite symmetric matrices under QR produce overlap matrices whose column structure cannot match the $|U_{e3}|/|U_{\mu 3}| \approx 0.17$ ratio required by PMNS.

[1] M. L. Gentry, Ckm quark mixing from geodesic axes of hyperbolic 3-manifold holonomy, Phys. Rev. D (2026), submitted March 2026, temporary ID es2026mar11_966.

[2] N. Cabibbo, Unitary symmetry and leptonic decays, Phys. Rev. Lett. **10**, 531 (1963).

[3] M. Kobayashi and T. Maskawa, Cp violation in the renormalizable theory of weak interaction, Prog. Theor. Phys. **49**, 652 (1973).

[4] B. Pontecorvo, Mesonium and anti-mesonium, Sov. Phys. JETP **6**, 429 (1957).

[5] Z. Maki, M. Nakagawa, and S. Sakata, Remarks on the unified model of elementary particles, Prog. Theor. Phys. **28**, 870 (1962).

[6] M. Culler, N. M. Dunfield, M. Goerner, and J. R. Weeks, SnapPy, a computer program for studying the geometry and topology of 3-manifolds (2024).

[7] P. D. Group, S. Navas, *et al.*, Review of particle physics, Phys. Rev. D **110**, 030001 (2024).